Muhammad Arslan Shaukat

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Work permit: § 16b AufenthG · 20 h/week in semester, full-time in breaks · no sponsorship needed



PROFILE

MSc Computer Science student at TU Dresden with two years shipping production backend systems on AWS. Built membership platforms end to end — Django and Flask services behind REST APIs, with automated email and document workflows across EC2, Lambda, S3 and RDS. Full-stack range in React, Next.js and Java/Spring Boot, with agentic tooling and AI-assisted development as a standing part of my workflow.

EDUCATION

Technische Universität Dresden *April 2026 – Expected March 2028*
MSc Computer Science · 1st Semester
Focus areas: Operating System Construction, Component Based Software Engineering, Data Analytics

Bahria University, Islamabad *March 2021 – February 2025*
BSc Computer Science · CGPA 3.38/4.0 (German Grade 1.9)
Focus areas: Artificial Intelligence, Natural Language Processing, Software Engineering, Web Engineering

PROFESSIONAL EXPERIENCE

RMCAANA Inc *Dec 2024 – May 2026*
Software Developer, Remote

- **Membership Platform:** Built the association's membership system from the ground up as a set of microservices on AWS, with a React front end over Django and Flask services exposed as RESTful APIs and backed by PostgreSQL on Amazon RDS. Automating registration and renewal took active membership from 250 to 400, and indexing and query tuning across PostgreSQL and MySQL kept member and payment lookups quick as the records grew.
- **Observership Portal:** Built the backend that carries an applicant through to placement, on Next.js 14 and PostgreSQL. RESTful service endpoints, conditional email routing through AWS SES and automated document storage on Amazon S3 moved applications, reviews and supporting files through the workflow without manual handling.
- **Microservices, Cloud and Delivery:** Designed and maintained the RESTful API layer tying the membership, observership and finance services together, keeping data consistent across distributed systems rather than copied between them by hand. Ran the whole estate on AWS with EC2 and Lambda for compute, Amazon S3 for documents, Amazon RDS for managed PostgreSQL and SES for transactional email, and added CI/CD pipelines so releases became repeatable instead of manual uploads.
- **Reliability and Testing:** Wrote Pytest and Unittest suites covering the registration, renewal and payment paths, and load tested the AWS deployment with K6, JMeter and AWS Distributed Load Testing before the annual renewal window, when most of the year's traffic arrives in a few days.
- **Partner Platforms and Financial Transparency:** Built and maintained web applications for RMCAANA Inc.'s joint initiatives with other organisations and delivered a secure portal where members and donors can pull up statements and records. Migrated legacy WordPress hosting onto AWS to cut costs and speed the site up.

Sybrid Pvt Ltd *May 2024 – Oct 2024*
Intern, Islamabad

- **Conversational AI:** Built Sybrid HRBot on Django and React with a RAG pipeline (ChatGPT API) that answered employee policy questions from internal documents, and developed the frontend for KissanBot, later launched as a commercial agricultural assistant.
- **AudioStream (Django, HLS.js):** Built a secure Django audio-streaming platform (HLS, .m3u8) so the team could listen to and transcribe proprietary voice recordings for the AgriVoice dataset without ever downloading the raw files — authentication-gated views and an HLS.js player enforced stream-only, leak-proof access.
- **Generative & Media AI:** Created Django tools for dataset labeling, and benchmarked Stable Diffusion (ToonCrafter) for storybook-to-animation workflows.

PROJECTS

- **4sight (Infineon Venture Hack 2026, Runner-up):** Built the Python backend for an agentic risk platform that surfaces operational project risks early. Jira Cloud data feeds a DAG that models how risk propagates between dependent tasks, and an MCP server exposes that graph as tools an LLM agent can call, with a ChromaDB vector store backing retrieval over project history. GitHub ↗
- **Kissan Voice:** Built an event-driven Spring Boot backend (Java 21, PostgreSQL, Kafka, AWS S3) with a transactional outbox for reliable event delivery and a resilient CRM integration that captures and manages voice-recording data end to end using retry, circuit breakers, and dead-letter queues. GitHub ↗
- **Operating System Construction (TU Dresden):** Booted my own 64-bit operating system on real hardware — freestanding C++ and x86-64 assembly, no C library and no heap: CGA and PS/2 drivers over memory-mapped and port I/O, IDT and 8259A PIC interrupt dispatch split into prologue/epilogue halves, a hand-written stack-switching context switch, PIT timer-driven preemptive scheduling and blocking semaphores. GitHub ↗
- **StockGenie:** Full-stack trading app with a Django and PostgreSQL backend that processes live market data concurrently through Redis and Celery, plus AI forecasting and FinBERT news sentiment. GitHub ↗

VOLUNTEERING

- **RMCAANA Volunteering (2024–2025):** Raised donations and helped procure surgical equipment for the Benazir Bhutto Hospital Urology Suite Upgrade in Rawalpindi, then helped channel flood-relief funds to the Al-Khidmat Foundation for 2025 Pakistan flood victims. Project details ↗
- **Stadt.Land.Oase, Haus der sozialen Vielfalt gGmbH (2026):** Idea developer in the Ideenlabor Dresden, an EU co-funded programme pairing young people with rural cultural organisations across Saxony on participatory community projects (conducted in German). Project details ↗

TECHNICAL SKILLS

Languages: Python, Java, C/C++, JavaScript, SQL
Backend & Web: Django, FastAPI, Flask, Spring Boot, Node.js, RESTful APIs, Kafka, Celery, Redis, React, Next.js
Databases & Cloud: PostgreSQL, MySQL, AWS (EC2, Lambda, S3, RDS, SES), Docker, CI/CD
AI & Agentic Systems: RAG pipelines, MCP servers and tool integrations, LLM APIs (OpenAI, Anthropic), agentic and DAG based workflows, vector stores (ChromaDB), n8n automation, Claude Code
Machine Learning: PyTorch, Scikit-learn, OpenCV, CUDA, FinBERT, Stable Diffusion
Testing & Tools: Pytest, Unittest, K6, JMeter, Git/GitHub, Linux

CERTIFICATIONS

- **Machine Learning Specialization** · Coursera (Stanford & DeepLearning.AI), Apr 2025
- **Introduction to CUDA C/C++** · NHR@FAU & NHR@TUD, Mar 2026 (Certificate of Attendance)
- **Google AI Essentials** · Coursera (Google), Mar 2025
- **Machine Learning Algorithms** · Great Learning Academy, Apr 2025
- **Introduction to Neural Networks and Deep Learning** · Great Learning Academy, Mar 2025
- **Fundamentals of Database Engineering** · Udemy, Mar 2025

AWARDS

- **1st Place, Project Gala 2025:** Bahria University, for an outstanding semester project.
- **1st Place, Speed Programming Competition 2022:** Bahria University.
- **Certificate of Achievement, NaSCon'23:** Fast NU.

LANGUAGES

English: C1, fluent in speech and writing
German: A2 and actively improving
Urdu / Punjabi: Native

INTERESTS

Reading · Gym & Fitness · Boxing · Robotics · Coding · Swimming · Music